

1. soru cevabı

$$f(x) = \ln(x-1) + \cos(x-1), \quad [1.3, 2] \text{ aralığında}$$

$$p_0 = 1.3$$

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}, \quad n \geq 1$$

$$f(x) = \ln(x-1) + \cos(x-1)$$

$$f'(x) = \frac{1}{(x-1)} - \sin(x-1) \Rightarrow f'(p_0) = f'(1.3)$$

$$= \frac{1}{1.3-1} - \sin(1.3-1) = 3.037813 \neq 0$$

$$\underline{a=1}$$

$$p_1 = p_0 - \frac{f(p_0)}{f'(p_0)} = 1.3 - \frac{(-0.248636)}{3.037813} = 1.381847$$

$$\begin{aligned} f(p_1) &= f(1.381847) = \ln(1.381847-1) + \cos(1.381847-1) \\ &= 0.034757 \end{aligned}$$

$$\begin{aligned} f'(p_1) &= f'(1.381847) \\ &= \frac{1}{1.381847-1} - \sin(1.381847-1) = 2.246214 \end{aligned}$$

1. Sorunun devamı

$$\underline{\alpha = 2}$$

$$p_2 = p_1 - \frac{f(p_1)}{f'(p_1)} = 1,381847 - \frac{(-0,034757)}{2,246214} = 1,3973205 \in [1,3,2]$$

$$f(p_2) = f(1,3973205) = \ln(1,3973205 - 1) + \cos(1,3973205 - 1) = 0,0009108$$

$$f'(p_2) = f'(1,3973205) = \frac{1}{1,3973205 - 1} - \sin(1,3973205 - 1) = 2,1299108$$

$$\underline{\alpha = 3}$$

$$p_3 = p_2 - \frac{f(p_2)}{f'(p_2)} = 1,3973205 - \frac{(-0,0009108)}{2,1299108} = 1,397748 \in [1,3,2]$$

$$f(p_3) = f(1,397748) = \ln(1,397748 - 1) + \cos(1,397748 - 1) = -0,000001$$

$$f'(p_3) = f'(1,397748) = \frac{1}{1,397748 - 1} - \sin(1,397748 - 1) = 2,126811$$

$$\underline{\alpha = 4}$$

$$p_4 = p_3 - \frac{f(p_3)}{f'(p_3)} = 1,3977484 - \frac{(-0,000001)}{2,126811} = 1,3977484 \in [1,3,2]$$

$$|p_4 - p_3| = |1,3977484 - 1,397748| = 0,0000004 = 0,4 \times 10^{-6}$$

$$\underline{\underline{\text{Kök} = 1,3977484}}$$

2. soru cevabı

$$f(x) = y$$

$$\Delta y_0 = \frac{1,59212 - 1,71828}{1,2 - 1,0} = 0,6308$$

$$\Delta y_1 = \frac{1,31119 - 1,59212}{1,4 - 1,2} = -1,40465$$

$$\Delta y_2 = \frac{0,85703 - 1,31119}{1,6 - 1,4} = -2,2708$$

$$\Delta y_3 = \frac{0,21765 - 0,85703}{1,8 - 1,6} = -3,1969$$

$$\Delta^2 y_0 = \frac{-1,40465 + 0,6308}{1,4 - 1,0} = -1,934625$$

$$\Delta^2 y_1 = \frac{-2,2708 + 1,40465}{1,6 - 1,2} = -2,16375$$

$$\Delta^2 y_2 = \frac{-3,1969 + 2,2708}{1,8 - 1,4} = -2,31525$$

$$\Delta^3 y_0 = \frac{-2,16375 + 1,934625}{1,6 - 1,0} = -0,384583333$$

$$\Delta^3 y_1 = \frac{-2,31525 + 2,16375}{1,8 - 1,2} = -0,249791666$$

$$\Delta^4 y_0 = \frac{-0,249791666 + 0,384583333}{1,8 - 1,0} = 0,168489583$$

$$f(1,25) = 1,7182 - 0,6308 \cdot (1,25 - 1,0) - 1,9346 \cdot (1,25 - 1,0) \cdot (1,25 - 1,2) - 0,3845 \cdot (1,25 - 1,0) \cdot (1,25 - 1,2) \cdot (1,25 - 1,4) + 0,1684 \cdot (1,25 - 1,0) \cdot (1,25 - 1,2) \cdot (1,25 - 1,4) \cdot (1,25 - 1,6) = 3,8508 \neq$$

$$\text{Gerçek Değer} = e^{1,25} - x^{1,25} = 3,4903 - 1,4531 = 1,5372$$

$$\text{Mutlak Hata} = \left| \frac{1,5372 - 3,8508}{1,5372} \right| = 5,4082 =$$

3. sorunun cevabı

$$f(x) = x - \sin x = 0,25$$

$$f(x) = x - \sin x - 0,25 = 0$$

$$x \in \left[0, \frac{\pi}{2}\right]$$

1. iterasyon $\Rightarrow x_0 = 0, x_1 = \frac{\pi}{2}$

$$|x_2 - x_1| < 10^{-4}$$

$$= -1,76 > 10^{-4}$$

$$f(x_0) = -0,25, f(x_1) = 0,32079$$

Devam

$$x_2 = x_0 - f(x_0) \cdot \frac{x_1 - x_0}{f(x_1) - f(x_0)} = 0,025 \cdot \frac{0,32079 - 0}{0,32079 - (-0,25)}$$

$$f(x_2) = -0,1969 =$$

$$x_2 = 0,6886 =$$

2. iterasyon

$$x_1 = \frac{\pi}{2}, x_2 = 0,6886$$

$$x_3 = x_1 - f(x_1) \cdot \frac{x_2 - x_1}{f(x_2) - f(x_1)} = \frac{\pi}{2} - 0,32 \cdot \frac{0,6886 - \frac{\pi}{2}}{-0,1969 - 0,32}$$

$$f(x_3) = -0,0801 =$$

$$x_3 = 1,0243 =$$

$$|x_3 - x_2| = 0,033110^{-4} \quad \text{Devam}$$

3. iterasyon

$$x_2 = 0,6886 \quad x_3 = 1,0243$$

$$x_4 = x_2 - f(x_2) \cdot \frac{x_3 - x_2}{f(x_3) - f(x_2)} = 0,6886 - (-0,1969) \cdot \frac{1,0243 - 0,6886}{-0,0801 - (-0,1969)} = 1,2544$$

$$f(x_4) = 0,054 =$$

$$x_4 = 1,2544 =$$

$$|x_4 - x_3| = 0,2301 > 10^{-4}$$

Devam

3. sonu devam

4. Iterasyon

$$x_3 = 1,0243 \quad x_4 = 1,2544$$

$$x_5 = x_3 - f(x_3) \cdot \frac{x_4 - x_3}{f(x_4) - f(x_3)} = 1,0243 - (-0,0801) \frac{1,2544 - 1,0243}{0,054 - (-0,0801)} = 1,1617$$

$$x_5 = 1,1617 \quad f(x_5) = -0,0058$$

$$|x_5 - x_4| = 0,0927 > 10^{-4} \text{ Devam}$$

5. Iterasyon

$$x_4 = 1,2544 \quad x_5 = 1,1617$$

$$x_6 = x_4 - f(x_4) \cdot \frac{x_5 - x_4}{f(x_5) - f(x_4)} = 1,2544 - 0,054 \frac{1,1617 - 1,2544}{-0,0058 - 0,054}$$

$$f(x_6) = -0,0003$$

$$x_6 = 1,1707$$

$$|x_6 - x_5| = 0,009 < 10^{-4}$$

$$Kök = 1,1707$$